



Photosynthesis in Higher Plants

Unit 4

PLANT PHYSIOLOGY - Unit Introduction

The description of the structure and variation of living organisms over a period of time ended up as **two apparently irreconcilable perspectives** on biology. The two rest on **two levels of organisation** of life forms and phenomena.

Perspective	Level of organisation described	Disciplines it resulted in
First	Life at the organismic and above level	Ecology and related disciplines
Second	Life at the cellular and molecular level	Physiology and biochemistry

In this unit the processes of **photosynthesis**, **respiration** and **plant growth and development** are described in molecular terms, but in the context of cellular activities and even at organism level. The relation of the physiological processes to environment is also discussed.

Profile

MELVIN CALVIN (1911-1997)

- **Melvin Calvin** was born in **Minnesota** in **April 1911**. He received his **Ph.D. in Chemistry** from the **University of Minnesota** and served as **Professor of Chemistry** at the **University of California, Berkeley**.
- Just after **world war II**, when the world was under shock after the **Hiroshima-Nagasaki bombings** and seeing the ill-effects of radio-activity, Calvin and co-workers put **radio-activity to beneficial use**.
- Calvin, along with **J.A. Bassham**, studied reactions in green plants forming sugar and other substances from raw materials like **carbon dioxide, water and minerals**, by **labelling the carbon dioxide with C¹⁴**.
- Calvin proposed that plants change **light energy to chemical energy** by transferring an electron in an **organised array of pigment molecules** and other substances.
- The **mapping of the pathway of carbon assimilation** in photosynthesis earned Melvin Calvin the **Nobel Prize in 1961**.
- The principles of photosynthesis as established by Calvin are at present being used in studies on **renewable resource for energy and materials** and basic studies in **solar energy research**.

Ch 11

PHOTOSYNTHESIS IN HIGHER PLANTS

All animals including human beings depend on plants for their food.

- **Autotrophs** - green plants make, or rather **synthesise**, the food they need through photosynthesis, and are therefore called autotrophs. **Autotrophic nutrition is found only in plants**; all other organisms that depend on the green plants for food are **heterotrophs**.
- **Photosynthesis** - a **physico-chemical process** by which green plants use **light energy** to drive the **synthesis of organic compounds**.

Ultimately, **all** living forms on earth depend on sunlight for energy. The use of energy from sunlight by plants doing photosynthesis is **the basis of life on earth**.

Photosynthesis is important due to **two reasons**:

- It is the **primary source of all food on earth**.
- It is also responsible for the **release of oxygen into the atmosphere** by green plants.

This chapter focusses on the **structure of the photosynthetic machinery** and the various reactions that transform **light energy into chemical energy**.

Simple experiments have shown that **chlorophyll** (green pigment of the leaf), **light** and **CO₂** are required for photosynthesis to occur.

During photosynthesis **carbon dioxide from the atmosphere is taken in by leaves through stomata** and used for making carbohydrates, **principally glucose and starch**.

Experiment 1 - is chlorophyll and light needed?

- 1 Two leaves are used: a **variegated leaf**, or a leaf that was **partially covered with black paper**.
- 2 Both are **exposed to light**.
- 3 The leaves are then **tested for the presence of starch**.
- 4 Result: photosynthesis occurred **only in the green parts of the leaves in the presence of light**.

Experiment 2 - is CO₂ needed?

- 1 A part of a leaf is enclosed in a **test tube containing some KOH soaked cotton**, which **absorbs CO₂**.
- 2 The **other half is exposed to air**.
- 3 The setup is placed **in light** for some time, then tested for starch.
- 4 Result: the **exposed part tested positive** for starch, while the portion inside the tube **tested negative** - showing that **CO₂ was required** for photosynthesis.

- In **1770** Joseph Priestley performed a series of experiments that revealed the **essential role of air** in the growth of green plants. Priestley **discovered oxygen in 1774**.

- 1 Priestley observed that a **candle burning in a closed space** - a **bell jar** - soon gets extinguished.
- 2 Similarly, a **mouse would soon suffocate** in a closed space.
- 3 He concluded that a burning candle or an animal that breathe the air **both somehow damage the air**.
- 4 When he placed a **mint plant** in the same bell jar, the **mouse stayed alive** and the **candle continued to burn**.

① **[NOTE]** Priestley's *hypothesis*: Plants restore to the air whatever breathing animals and burning candles remove.

- Using a **similar setup** as the one used by Priestley, but by placing it **once in the dark and once in the sunlight**, Ingenhousz showed that **sunlight is essential** to the plant process that somehow **purifies the air** fouled by burning candles or breathing animals.
- In an **elegant experiment with an aquatic plant** he showed that in **bright sunlight small bubbles** were formed around the **green parts**, while **in the dark they did not**. He later identified these bubbles to be of **oxygen**. He hence showed that it is **only the green part** of the plants that could release oxygen.

- It was not until about **1854** that **Julius von Sachs** provided evidence for the **production of glucose** when plants grow. Glucose is **usually stored as starch**. Sachs found that the **green parts in plants** is where **glucose is made**, and that the glucose is usually stored as starch.
- His later studies showed that the green substance in plants (chlorophyll) is located in special bodies, later called chloroplasts, within plant cells.

11.2d

T.W Engelmann (1843-1909)

1

Using a **prism**, Engelmann **split light into its spectral components**.

2

He then **illuminated a green alga, *Cladophora***, placed in a **suspension of aerobic bacteria**.

3

The **bacteria were used to detect the sites of O_2 evolution**.

4

The bacteria **accumulated mainly in the region of blue and red light** of the split spectrum.

5

A **first action spectrum of photosynthesis** was thus described; it resembles roughly the **absorption spectra of chlorophyll a and b**.

11.2e

The First Empirical Equation

- By the **middle of the nineteenth century** the key features of plant photosynthesis were known, namely that plants could use **light energy to make carbohydrates from CO_2 and water**.
- The **empirical equation** representing the total process of photosynthesis for **oxygen evolving organisms** was then understood as: $CO_2 + H_2O$, in the presence of **light**, yields $[CH_2O] + O_2$. Here $[CH_2O]$ represented a **carbohydrate** (e.g., **glucose, a six-carbon sugar**).

11.2f

Cornelius van Niel (1897-1985)

- A **milestone contribution** to the understanding of photosynthesis was made by a **microbiologist, Cornelius van Niel**. Based on his studies of **purple and green bacteria**, he demonstrated that photosynthesis is essentially a **light-dependent reaction** in which **hydrogen from a suitable oxidisable compound reduces carbon dioxide to carbohydrates**.
- His finding can be expressed by: $2H_2A + CO_2$, in the presence of **light**, yields $2A + CH_2O + H_2O$.
- In **green plants H_2O is the hydrogen donor** and is **oxidised to O_2** . **Some organisms do not release O_2 during photosynthesis**: when H_2S instead is the hydrogen donor for **purple and green sulphur bacteria**, the 'oxidation' product is **sulphur or sulphate** depending on the organism, and **not O_2** .
- Van Niel hence inferred that the **O_2 evolved by the green plant comes from H_2O , not from carbon dioxide**. This was later **proved by using radioisotopic techniques** - the O_2 released is **from water**.
- The **correct equation** that would represent the overall process of photosynthesis is: $6CO_2 + 12H_2O$, in the presence of **light**, yields $C_6H_{12}O_6 + 6H_2O + 6O_2$, where $C_6H_{12}O_6$ represents **glucose**. This is **not a single reaction** but a description of a **multistep process** called photosynthesis.

11.3

WHERE DOES PHOTOSYNTHESIS TAKE PLACE?

You would of course answer: in '**the green leaf**' or '**in the chloroplasts**', based on what you earlier read in Chapter 8.

- Photosynthesis **does** take place in the **green leaves** of plants, but it does so **also in other green parts** of the plants.
- The **mesophyll cells** in the leaves have a **large number of chloroplasts**. **Usually the chloroplasts align themselves along the walls of the mesophyll cells**, such that they get the **optimum quantity of the incident light**.
- Photosynthesis has **two stages**: the **light reaction** and the **carbon fixing reactions**.

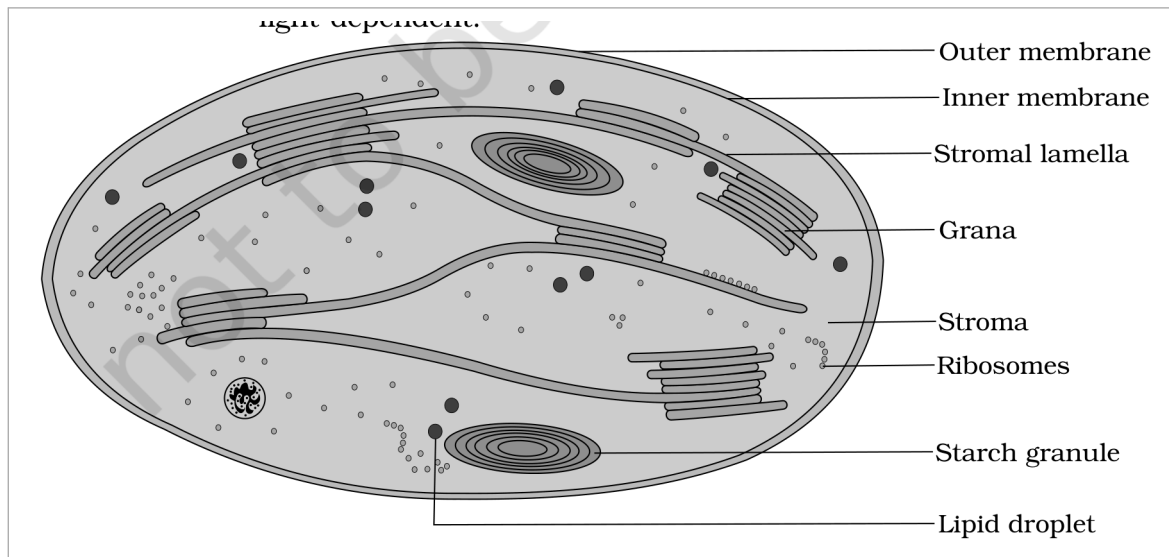


Fig. 11.2 - Diagrammatic representation of an electron micrograph of a section of chloroplast.

Within the chloroplast there is a **membranous system** consisting of the **grana**, the **stroma lamellae**, and the **matrix stroma**. The parts marked on the section of the chloroplast are:

Part of the chloroplast	What it is
Outer membrane	The outer of the two bounding membranes of the chloroplast
Inner membrane	The inner bounding membrane, lying within the outer membrane
Grana	The stacked membrane compartments of the membranous system
Stromal lamella	The membrane lamellae connecting the grana; also written stroma lamellae
Stroma	The matrix in which the grana and stromal lamella are embedded
Ribosomes	Present in the stroma of the chloroplast
Starch granule	Stored product lying in the stroma
Lipid droplet	Lipid body lying in the stroma

There is a clear **division of labour** within the chloroplast:

- The **membrane system** is responsible for **trapping the light energy** and also for the **synthesis of ATP and NADPH**.
- In **stroma**, **enzymatic reactions synthesise sugar**, which in turn forms **starch**.
- Within the chloroplasts the **membranes are sites for the light reaction**, while the **chemosynthetic pathway occurs in the stroma**.
- **Light reactions** (photochemical reactions) - the reactions of the membrane system, since they are **directly light driven**.
- **Dark reactions** (carbon reactions) - the **stromal reactions**. They are **not directly light driven** but are **dependent on the products of light reactions (ATP and NADPH)**; hence, to distinguish them, they are called dark reactions **by convention**.

① **[NOTE]** That the stromal reactions are called **dark reactions** should not be construed to mean that they **occur in darkness** or that they are **not light-dependent**.

Leaves differ in shade of green because they contain several pigments—chlorophyll a, chlorophyll b, xanthophylls and carotenoids—in different proportions.

- The leaf pigments of any green plant can be separated through **paper chromatography**. A chromatographic separation shows that the colour that we see in leaves is **not due to a single pigment but due to four pigments**.

Leaf pigment	Colour in the chromatogram
Chlorophyll a	Bright or blue green
Chlorophyll b	Yellow green
Xanthophylls	Yellow
Carotenoids	Yellow to yellow-orange

- **Pigments** - substances that have an **ability to absorb light at specific wavelengths**.

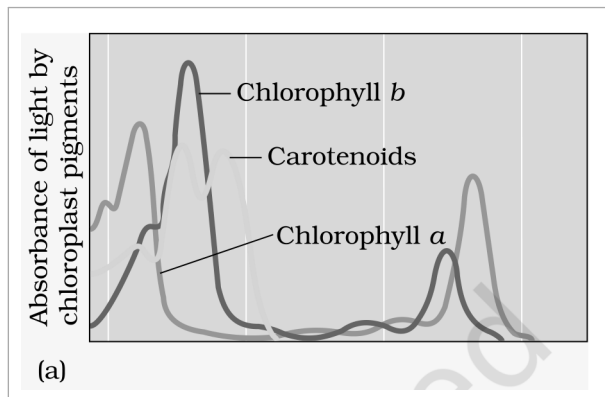


Fig. 11.3a - Graph showing the **absorption spectrum** of chlorophyll a, b and the carotenoids. Its vertical axis is **Absorbance of light by chloroplast pigments**, and the curves - separated by colour in the original - are labelled **Chlorophyll a**, **Chlorophyll b** and **Carotenoids**. Panel marker: **(a)**.

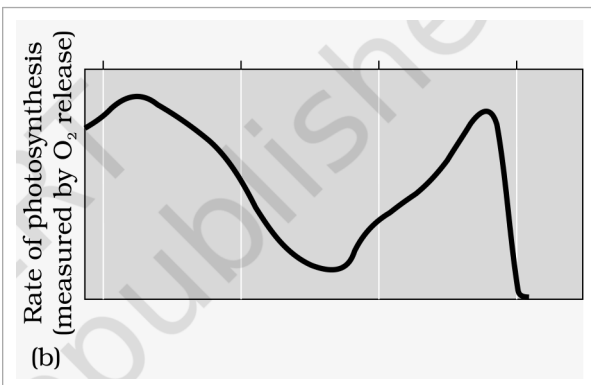


Fig. 11.3b - Graph showing the **action spectrum of photosynthesis**. Its vertical axis is **Rate of photosynthesis (measured by O₂ release)**. Panel marker: **(b)**.

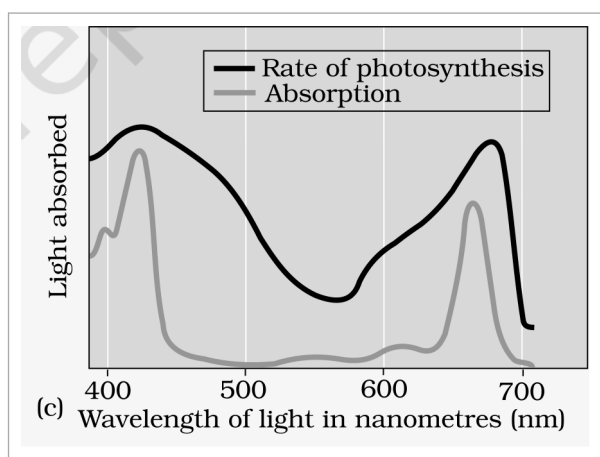


Fig. 11.3c - Graph showing the **action spectrum of photosynthesis superimposed on the absorption spectrum of chlorophyll a**. The original separated the two curves by colour: the **Rate of photosynthesis** curve was black and the **Absorption** curve was cyan, and the shaded band is **Light absorbed**. Its horizontal axis is **Wavelength of light in nanometres (nm)**, marked at 400, 500, 600 and 700. Panel marker: **(c)**.

- The wavelengths at which there is **maximum absorption by chlorophyll a**, i.e., in the **blue and the red regions**, also show a **higher rate of photosynthesis**. Hence we can conclude that **chlorophyll a is the chief pigment associated with photosynthesis**.
- There is **not a complete one-to-one overlap** between the **absorption spectrum of chlorophyll a** and the **action spectrum of photosynthesis**. The graphs together show that **most of the photosynthesis takes place in the blue and red regions of the spectrum**; **some photosynthesis does take place at the other wavelengths of the visible spectrum**.
- **Accessory pigments** - though chlorophyll is the **major pigment** responsible for trapping light, other thylakoid pigments like **chlorophyll b**, **xanthophylls** and **carotenoids** are called accessory pigments. They also **absorb light and transfer the energy to chlorophyll a**.

- The accessory pigments **not only** enable a **wider range of wavelength** of incoming light to be utilised for photosynthesis, **but also protect chlorophyll a from photo-oxidation**.

11.5

WHAT IS LIGHT REACTION?

- **Light reactions** or the '**Photochemical**' phase include **light absorption, water splitting, oxygen release**, and the **formation of high-energy chemical intermediates, ATP and NADPH**.
- **Several protein complexes** are involved in the light reaction. The pigments are organised into **two discrete photochemical light harvesting complexes (LHC)** within **Photosystem I (PS I)** and **Photosystem II (PS II)**.

① **[NOTE]** The photosystems are named **in the sequence of their discovery**, and **not** in the sequence in which they **function** during the light reaction.

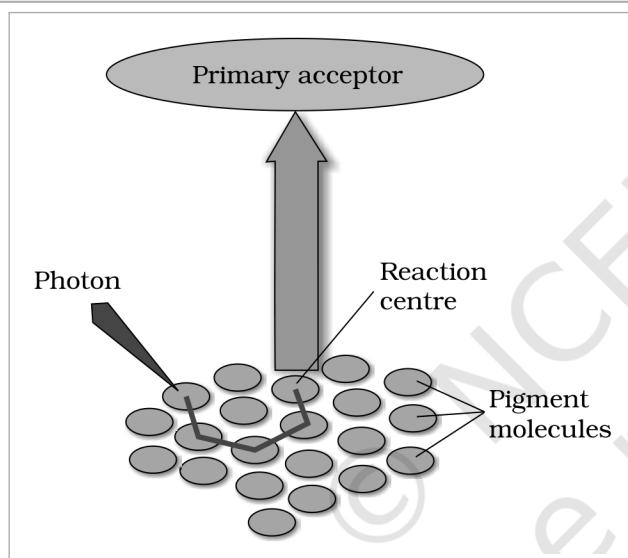


Fig. 11.4 - The light harvesting complex.

The parts marked on the light harvesting complex are the **Pigment molecules**, the **Reaction centre**, the **Primary acceptor**, and the incoming **Photon** of light.

- The **LHC** are made up of **hundreds of pigment molecules** bound to **proteins**.
- **Each** photosystem has **all** the pigments (**except one molecule of chlorophyll a**) forming a **light harvesting system**, also called **antennae**. These **antenna pigments** help to make photosynthesis **more efficient** by absorbing **different wavelengths of light**.
- The **single chlorophyll a molecule** forms the **reaction centre**, and the reaction centre is **different in both the photosystems**.
- In the light reaction the light energy is **absorbed by the pigments present in the antenna** and **funnelled to special chlorophyll a molecules** called **reaction centre chlorophylls**.

Photosystem	Reaction centre chlorophyll a	Absorption peak	Called
PS I	One chlorophyll a molecule	700 nm	P700
PS II	One chlorophyll a molecule	680 nm	P680

11.6

THE ELECTRON TRANSPORT

- 1 In **photosystem II** the reaction centre chlorophyll a **absorbs 680 nm wavelength of red light**, causing **electrons to become excited** and **jump into an orbit farther from the atomic nucleus**.
- 2 These electrons are **picked up by an electron acceptor**, which passes them to an **electron transport system consisting of cytochromes**.
- 3 This movement of electrons is **downhill**, in terms of an **oxidation-reduction or redox potential scale**. The electrons are **not used up** as they pass through the electron transport chain, but are **passed on to the pigments of photosystem PS I**.

4

Simultaneously, electrons in the **reaction centre of PS I** are also excited when they receive **red light of wavelength 700 nm**, and are transferred to **another acceptor molecule that has a greater redox potential**.

5

These electrons then are **moved downhill again**, this time to a molecule of **energy-rich NADP^+** . The addition of these electrons **reduces NADP^+ to $\text{NADPH} + \text{H}^+$** .

- **Z scheme** - the whole scheme of transfer of electrons, starting from **PS II, uphill to the acceptor, down the electron transport chain to PS I, excitation of electrons, transfer to another acceptor, and finally downhill to NADP^+ reducing it to $\text{NADPH} + \text{H}^+$** . It is so called due to its **characteristic shape**, which is formed when **all the carriers are placed in a sequence on a redox potential scale**.

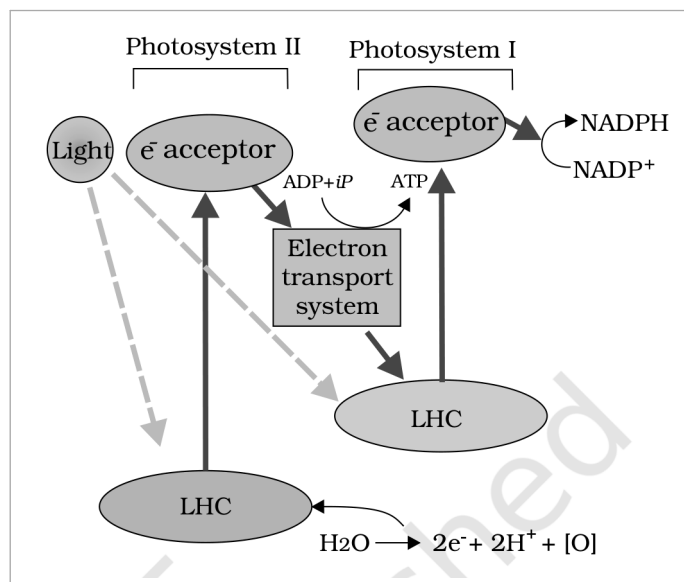


Fig. 11.5 - Z scheme of light reaction.

The Z scheme diagram is marked with **Photosystem II** and **Photosystem I**, each receiving **Light** through its **LHC**; the **e acceptor** above each photosystem; the **Electron transport system** joining them; **ADP + iP** being converted to **ATP**; **NADP^+** being reduced to **NADPH** ; and, at PS II, **H_2O** being split to give **$2\text{e}^- + 2\text{H}^+ + [\text{O}]$** .

11.6.1

Splitting of Water

Photosystem II receives replacement electrons continuously through the splitting of water (photolysis).

- The electrons that were **moved from photosystem II must be replaced**, and this is achieved by **electrons available due to splitting of water**.
- The **splitting of water is associated with the PS II**; water is split into **2H^+ , $[\text{O}]$ and electrons**. This creates **oxygen**, one of the **net products of photosynthesis**.
- The electrons needed to **replace those removed from photosystem I** are provided by **photosystem II**.
- The water-splitting reaction is: **$2\text{H}_2\text{O}$ yields $4\text{H}^+ + \text{O}_2 + 4\text{e}^-$** .

① **[NOTE]** The **water splitting complex** is associated with the **PS II**, which itself is **physically located on the inner side of the membrane of the thylakoid**.

11.6.2

Cyclic and Non-cyclic Photo-phosphorylation

Living organisms have the capability of **extracting energy from oxidisable substances** and **store this in the form of bond energy**. Special substances like **ATP** carry this energy in their **chemical bonds**.

- **Phosphorylation** - the process through which **ATP is synthesised by cells (in mitochondria and chloroplasts)**.
- **Photophosphorylation** - the **synthesis of ATP from ADP and inorganic phosphate in the presence of light**.
- **Non-cyclic photo-phosphorylation** - occurs when the **two photosystems work in a series, first PS II and then the PS I**. The two photosystems are connected through an **electron transport chain**, as seen in the **Z scheme**, and **both ATP and $\text{NADPH} + \text{H}^+$** are synthesised by this kind of electron flow.

- When **only PS I is functional**, the electron is **circulated within the photosystem** and the phosphorylation occurs due to **cyclic flow of electrons**. A **possible location** where cyclic photophosphorylation could be happening is in the **stroma lamellae**.
- While the **membrane or lamellae of the grana** have **both PS I and PS II**, the **stroma lamellae membranes** **lack PS II** as well as **NADP reductase enzyme**.

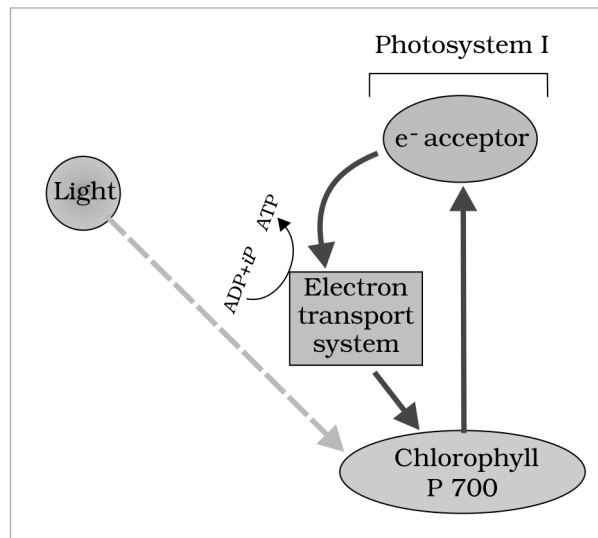


Fig. 11.6 - Cyclic photophosphorylation.

The cyclic photophosphorylation diagram is marked with **Photosystem I** alone, its incoming **Light**, its **Chlorophyll P 700** reaction centre, the **e⁻ acceptor** above it, the **Electron transport system** returning the electron to the same photosystem, and **ADP + iP** being converted to **ATP**.

- In **cyclic flow** the excited electron **does not pass on to NADP⁺** but is **cycled back to the PS I complex** through the electron transport chain. The cyclic flow hence results **only in the synthesis of ATP, but not of NADPH + H⁺**.
- Cyclic photophosphorylation **also** occurs when **only light of wavelengths beyond 680 nm** are available for excitation.

11.6.3 Chemiosmotic Hypothesis

ATP synthesis in the chloroplast is explained by the chemiosmotic hypothesis.

- **Like in respiration**, in photosynthesis too **ATP synthesis is linked to development of a proton gradient across a membrane**; this time these are the **membranes of thylakoid**.
- There is **one difference** though: here the **proton accumulation is towards the inside of the membrane**, i.e., in the **lumen**, whereas in **respiration** protons accumulate in the **intermembrane space of the mitochondria** when electrons move through the ETS.

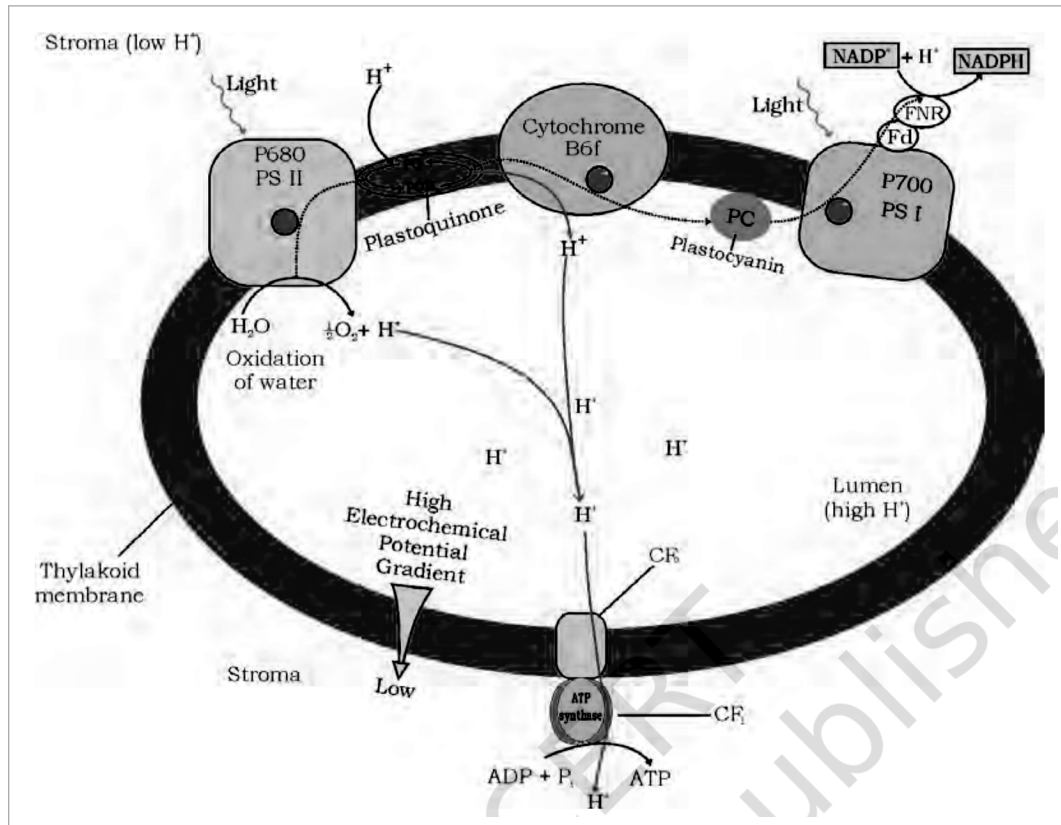


Fig. 11.7 - ATP synthesis through chemiosmosis.

The chemiosmosis diagram is drawn across the **Thylakoid membrane**, with the **Stroma (low H^+)** on the outer side and the **Lumen (high H^+)** on the inner side. Along the membrane, in order, it is marked: incoming **Light** at **P680 PS II**, then **Plastoquinone**, **Cytochrome B6f**, **PC (Plastocyanin)**, then **Light** again at **P700 PS I**, then **Fd** and **FNR**, which reduces $NADP^+ + H^+$ to **NADPH**. On the lumen side the **Oxidation of water** splits H_2O into $\frac{1}{2}O_2 + H^+$, releasing H^+ into the lumen.

The vertical scale beside the membrane runs from **High Electrochemical Potential Gradient** down to **Low**, from the **Lumen** to the **Stroma**. At the right the **ATP synthase** is drawn in its two parts, **CF₀** in the membrane and **CF₁** facing the stroma, converting **ADP + Pi** into **ATP**.

- 1 Since **splitting of the water molecule** takes place on the **inner side of the membrane**, the **protons or hydrogen ions** that are produced by the splitting of water **accumulate within the lumen of the thylakoids**.
 - 2 As electrons move through the photosystems, **protons are transported across the membrane**, because the **primary acceptor of electron**, which is located **towards the outer side of the membrane**, transfers its electron **not to an electron carrier but to an H carrier**.
 - 3 Hence this **H-carrier molecule removes a proton from the stroma** while transporting an electron; when this molecule **passes on its electron to the electron carrier on the inner side of the membrane**, the **proton is released into the inner side or the lumen side** of the membrane.
 - 4 The **NADP reductase enzyme** is located on the **stroma side of the membrane**. Along with electrons that come from the **acceptor of electrons of PS I**, **protons are necessary for the reduction of $NADP^+$ to $NADPH + H^+$** ; these protons are **also removed from the stroma**.
 - 5 Hence, within the chloroplast, **protons in the stroma decrease in number**, while **in the lumen there is accumulation of protons**. This creates a **proton gradient across the thylakoid membrane** as well as a **measurable decrease in pH in the lumen**.
- The **proton gradient** is important because it is the **breakdown of this gradient** that leads to the **synthesis of ATP**. The gradient is broken down due to the **movement of protons across the membrane to the stroma** through the **transmembrane channel of the CF₀ of the ATP synthase**.

- **ATP synthase** - the enzyme consists of **two parts**. One, called CF_0 , is **embedded in the thylakoid membrane** and forms a **transmembrane channel** that carries out **facilitated diffusion of protons across the membrane**. The other portion is called CF_1 and **protrudes on the outer surface of the thylakoid membrane on the side that faces the stroma**.
- The breakdown of the gradient provides **enough energy to cause a conformational change in the CF_1 particle** of the ATP synthase, which makes the enzyme **synthesise several molecules of energy-packed ATP**.

Chemiosmosis requires:

- a **membrane**;
- a **proton pump**;
- a **proton gradient**;
- **ATP synthase**.

Energy is used to pump protons across a membrane, to create a **gradient or a high concentration of protons within the thylakoid lumen**. ATP synthase has a **channel that allows diffusion of protons back across the membrane**, and this **releases enough energy to activate ATP synthase enzyme** that catalyses the formation of ATP.

- Along with the **NADPH** produced by the movement of electrons, the **ATP will be used immediately in the biosynthetic reaction taking place in the stroma**, responsible for **fixing CO_2 and synthesis of sugars**.

11.7

WHERE ARE THE ATP AND NADPH USED?

- The **products of light reaction** are **ATP, NADPH and O_2** . Of these, **O_2 diffuses out of the chloroplast**, while **ATP and NADPH are used to drive the processes leading to the synthesis of food, more accurately sugars**.
- **Biosynthetic phase of photosynthesis** - the **synthesis of sugars using ATP and NADPH**.
- The biosynthetic process **does not directly depend on the presence of light** but is **dependent on the products of the light reaction**, i.e., **ATP and NADPH**, besides **CO_2 and H_2O** .
- **Immediately after light becomes unavailable** the biosynthetic process **continues for some time and then stops**. If then **light is made available**, the **synthesis starts again**.
- **CO_2 is combined with H_2O to produce $(CH_2O)_n$ or sugars**. The use of **radioactive ^{14}C by Melvin Calvin** in algal photosynthesis studies led to the discovery that the **first CO_2 fixation product was a 3-carbon organic acid**. Calvin also contributed to working out the **complete biosynthetic pathway**; hence it was called the **Calvin cycle** after him. The **first product identified was 3-phosphoglyceric acid**, or in short **PGA**.
- Experiments conducted over a **wide range of plants** led to the discovery of **another group of plants** where the **first stable product of CO_2 fixation was again an organic acid**, but one which had **4 carbon atoms** in it. This acid was identified to be **oxaloacetic acid** or **OAA**.

Type of CO_2 assimilation	First product of CO_2 fixation	Pathway
Plants whose first product is a C_3 acid	PGA	the C_3 pathway
Plants whose first product is a C_4 acid	OAA	the C_4 pathway

11.7.1

The Primary Acceptor of CO_2

The dark reaction involves two key points: the primary CO_2 acceptor and the first stable product of CO_2 fixation.

- Scientists believed that **since the first product was a C_3 acid**, the **primary acceptor would be a 2-carbon compound**. They spent **many years** trying to identify a 2-carbon compound before they discovered the 5-carbon RuBP.
- The studies **very unexpectedly** showed that the acceptor molecule was a **5-carbon ketose sugar - ribulose biphosphate (RuBP)**.

- **Calvin and his co-workers** then worked out the **whole pathway** and showed that the pathway **operated in a cyclic manner**; the **RuBP was regenerated**.

① **[NOTE]** The Calvin pathway occurs in all photosynthetic plants; it does not matter whether they have C_3 or C_4 (or any other) pathways.

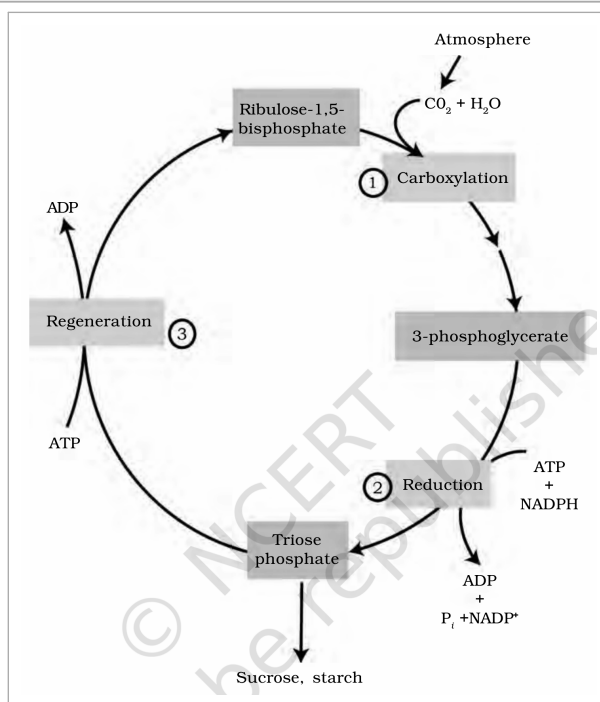


Fig. 11.8 - The Calvin cycle proceeds in three stages: (1) **carboxylation**, during which CO_2 combines with ribulose-1,5-bisphosphate; (2) **reduction**, during which carbohydrate is formed at the expense of the photochemically made ATP and NADPH; and (3) **regeneration** during which the CO_2 acceptor ribulose-1,5-bisphosphate is formed again so that the cycle continues.

The Calvin cycle diagram is marked with $CO_2 + H_2O$ entering from the **Atmosphere**; the three numbered stages **1, 2 and 3 - Carboxylation, Reduction and Regeneration**; the intermediates **Ribulose-1,5-bisphosphate, 3-phosphoglycerate and Triose phosphate**; the cofactors **ATP and ADP** on the regeneration step and **ATP + NADPH** yielding **ADP + P_i + $NADP^+$** on the reduction step; and the products drawn leaving the cycle as **Sucrose, starch**.

For ease of understanding, the Calvin cycle can be described under **three stages: carboxylation, reduction and regeneration**.

Stage 1

Carboxylation

- **Carboxylation** is the **fixation of CO_2 into a stable organic intermediate**. It is the **most crucial step of the Calvin cycle**, where CO_2 is utilised for the carboxylation of RuBP.
- The reaction is **catalysed by the enzyme RuBP carboxylase**, which results in the **formation of two molecules of 3-PGA**.
- **RuBisCO** - since the **RuBP carboxylase enzyme also has an oxygenation activity**, it would be **more correct** to call it **RuBP carboxylase-oxygenase**, or RuBisCO.

Stage 2

Reduction

- **Reduction** is a **series of reactions that lead to the formation of glucose**. The steps involve **utilisation of 2 molecules of ATP for phosphorylation and two of NADPH for reduction, per CO_2 molecule fixed**.
- The **fixation of six molecules of CO_2 and 6 turns of the cycle** are required for the **formation of one molecule of glucose** from the pathway.

- **Regeneration of the CO_2 acceptor molecule RuBP is crucial if the cycle is to continue uninterrupted.** The regeneration steps require **one ATP for phosphorylation to form RuBP.**
- Hence **for every CO_2 molecule entering the Calvin cycle, 3 molecules of ATP and 2 of NADPH are required.** It is **probably** to meet this **difference in number of ATP and NADPH used in the dark reaction** that the **cyclic phosphorylation** takes place.

Calvin cycle balance sheet	IN	OUT
Carbon dioxide	six CO_2	-
Energy carrier	18 ATP	18 ADP
Reducing power	12 NADPH	12 NADP
Sugar	-	one glucose

☆ **[MEMORY AID - not in NCERT]** For the Calvin cycle stages, remember **C-R-R**: Carboxylation, Reduction, Regeneration. And for the per- CO_2 cost, "**3 A, 2 N**" - 3 ATP and 2 NADPH.

11.8 THE C_4 PATHWAY

- **Plants that are adapted to dry tropical regions have the C_4 pathway.** Though C_4 plants have the C_4 oxaloacetic acid as the first CO_2 fixation product, they use the C_3 pathway or the Calvin cycle as the main biosynthetic pathway.

C_4 plants are special - they:

- have a **special type of leaf anatomy**;
- **tolerate higher temperatures**;
- show a **response to high light intensities**;
- **lack a process called photorespiration**;
- have **greater productivity of biomass**.

- **Bundle sheath cells** - the **particularly large cells around the vascular bundles** of the C_4 plants. Leaves which have such anatomy are said to have '**Kranz**' anatomy. '**Kranz**' means '**wreath**' and is a **reflection of the arrangement of cells**.

The **bundle sheath cells may form several layers** around the vascular bundles. They are characterised by having:

- a **large number of chloroplasts**;
- **thick walls impervious to gaseous exchange**;
- **no intercellular spaces**.
- **Maize and sorghum** are C_4 plants whose **leaves may be sectioned** to observe the **Kranz anatomy** and the **distribution of mesophyll cells**. The **presence of the bundle sheath around the vascular bundles** would help you **identify the C_4 plants**.

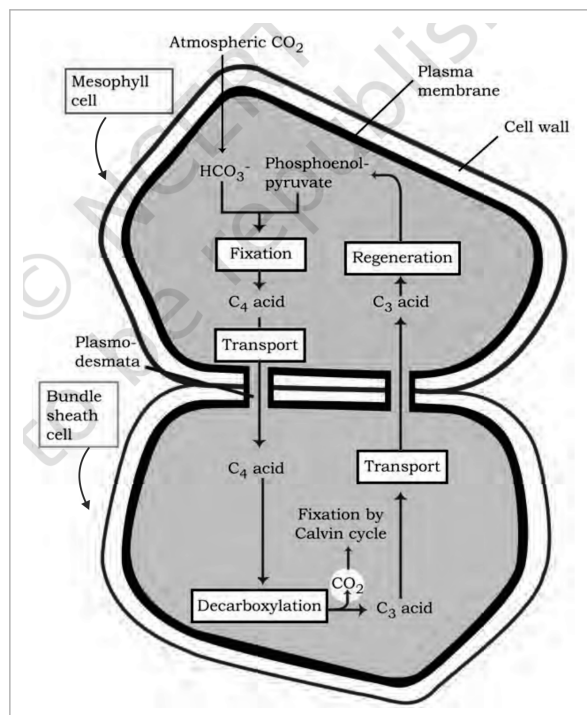


Fig. 11.9 - Diagrammatic representation of the Hatch and Slack Pathway.

The Hatch and Slack Pathway diagram is drawn across two cells. In the **Mesophyll cell**, bounded by its **Cell wall** and **Plasma membrane**, **Atmospheric CO₂** enters as **HCO₃⁻** and undergoes **Fixation** with **Phosphoenol-pyruvate** to give a **C₄ acid**. The **C₄ acid** undergoes **Transport** through the **Plasmodesmata** into the **Bundle sheath cell**, where **Decarboxylation** releases **CO₂** for **Fixation by Calvin cycle**, and the **C₃ acid** returns to the mesophyll cell for **Regeneration** of phosphoenol-pyruvate.

- The **C₄** pathway shown in Figure 11.9 has been named the **Hatch and Slack Pathway**, and is again a **cyclic process**.
- The **primary CO₂ acceptor** is a **3-carbon molecule phosphoenol pyruvate (PEP)** and is **present in the mesophyll cells**. The enzyme responsible for the **primary CO₂ fixation** in **C₄ plants** is **PEP carboxylase** or **PEPcase**.

① **[NOTE]** Mesophyll cells of **C₄ plants** lack **RuBisCO**.

- (cycle - last step feeds back to step 1)
- ▲ 1 The **C₄ acid OAA** is formed in the mesophyll cells.
 - ▲ 2 OAA then forms **other 4-carbon compounds like malic acid or aspartic acid** in the **mesophyll cells itself**, which are **transported to the bundle sheath cells**.
 - ▲ 3 In the **bundle sheath cells** these **C₄ acids** are broken down to release **CO₂** and a **3-carbon molecule**.
 - ▲ 4 The **3-carbon molecule** is transported back to the mesophyll, where it is **converted to PEP again**, thus **completing the cycle**.
 - ▲ 5 The **CO₂** released in the bundle sheath cells enters the **C₃ or the Calvin pathway**, a pathway common to all plants.
- The **bundle sheath cells** are rich in an enzyme **Ribulose biphosphate carboxylase-oxygenase (RuBisCO)**, but **lack PEPcase**.
 - Thus the **basic pathway that results in the formation of the sugars**, the **Calvin pathway**, is **common to the C₃ and C₄ plants**. It occurs in **all the mesophyll cells of the C₃ plants**; in the **C₄ plants** it does not take place in the mesophyll cells but does so **only in the bundle sheath cells**.

Kranz anatomy is an **internal** criterion. Whether a plant can be told apart as C_3 or C_4 by looking at it **externally** is answered in the closing section, 'Terms used in the exercises'.

11.9

PHOTORESPIRATION

Photorespiration is one more process that creates an important difference between C_3 and C_4 plants.

- The first CO_2 fixation step of the Calvin pathway is the reaction where **RuBP combines with CO_2 to form 2 molecules of 3PGA**, catalysed by **RuBisCO**: $\text{RuBP} + \text{CO}_2$, catalysed by **RuBisCO**, yields $2 \times 3\text{PGA}$.
- **RuBisCO is the most abundant enzyme in the world.** It is characterised by the fact that its **active site can bind to both CO_2 and O_2** - hence the name.
- RuBisCO has a **much greater affinity for CO_2** when the $\text{CO}_2:\text{O}_2$ is **nearly equal**. The binding of CO_2 and O_2 to RuBisCO is **competitive**, and it is the **relative concentration of O_2 and CO_2** that determines **which of the two will bind** to the enzyme.
- In C_3 plants some O_2 does bind to RuBisCO, and hence CO_2 fixation is decreased.
- **Photorespiration** - here the **RuBP, instead of being converted to 2 molecules of PGA, binds with O_2 to form one molecule of phosphoglycerate and phosphoglycolate (2 Carbon).**
- In the **photorespiratory pathway** there is **neither synthesis of sugars nor of ATP**; rather it results in the **release of CO_2 with the utilisation of ATP**. There is **no synthesis of ATP or NADPH**.

① [NOTE] The biological function of photorespiration is not known yet.

- **RuBisCO also catalyses a wasteful oxygenation reaction in C_3 plants:** photorespiration.
- In C_4 plants **photorespiration does not occur**, because they have a **mechanism that increases the concentration of CO_2 at the enzyme site**.
- This takes place when the **C_4 acid from the mesophyll is broken down in the bundle sheath cells to release CO_2** , which results in **increasing the intracellular concentration of CO_2** . In turn this ensures that the **RuBisCO functions as a carboxylase, minimising the oxygenase activity**.
- Since the **C_4 plants lack photorespiration, productivity and yields are better** in these plants. In addition, these plants show **tolerance to higher temperatures**.

Table 11.1

DIFFERENCES BETWEEN C_3 AND C_4 PLANTS

NCERT prints **TABLE 11.1** with **Columns 2 and 3 left blank**, to be filled in to highlight the **differences between C_3 and C_4 Plants**. The completed table, filled from this chapter's own statements, is:

Characteristic	C_3 plants	C_4 plants
Cell type in which the Calvin cycle takes place	Mesophyll	Bundle sheath
Cell type in which the initial carboxylation reaction occurs	Mesophyll	Mesophyll
How many cell types the leaf has that fix CO_2	One: Mesophyll	Two: Bundle sheath and mesophyll
Which is the primary CO_2 acceptor	RuBP	PEP
Number of carbons in the primary CO_2 acceptor	5	3
Which is the primary CO_2 fixation product	PGA	OAA
Number of carbons in the primary CO_2 fixation product	3	4
Does the plant have RuBisCO	Yes	Yes
Does the plant have PEP Case	No	Yes
Which cells in the plant have RuBisCO	Mesophyll	Bundle sheath
CO_2 fixation rate under high light conditions	Low	High
Is photorespiration present at low light intensities	Negligible	Negligible
Is photorespiration present at high light intensities	High	Negligible
Would photorespiration be present at low CO_2 concentrations	High	Negligible

Characteristic	C ₃ plants	C ₄ plants
Would photorespiration be present at high CO₂ concentrations	Negligible	Negligible
Temperature optimum	20-25 degrees C	30-40 degrees C
Examples	Tomatoes, bell pepper	Maize, sorghum

The 'Choose from' options NCERT supplies with the table are: **Mesophyll / Bundle sheath / both**; **Two: Bundle sheath and mesophyll**, **One: Mesophyll**, **Three: Bundle sheath, palisade, spongy mesophyll**; **RuBP / PEP / PGA**; **5 / 4 / 3**; **PGA / OAA / RuBP / PEP**; **3 / 4 / 5**; **Yes / No / Not always**; **Mesophyll / Bundle sheath / none**; **Low / high / medium**; **High / negligible / sometimes**; and **30-40 degrees C / 20-25 degrees C / above 40 degrees C**.

11.10

FACTORS AFFECTING PHOTOSYNTHESIS

An understanding of the **factors that affect photosynthesis** is necessary, because the **rate of photosynthesis is very important in determining the yield of plants**, including **crop plants**.

Photosynthesis is under the influence of **several factors, both internal (plant) and external**:

Factor group	Factors it includes
Plant (internal) factors	The number, size, age and orientation of leaves , mesophyll cells and chloroplasts , internal CO₂ concentration , and the amount of chlorophyll . These are dependent on the genetic predisposition and the growth of the plant .
External factors	The availability of sunlight, temperature, CO₂ concentration and water .

- As a plant photosynthesises, **all these factors will simultaneously affect its rate**. Hence, though **several factors interact and simultaneously affect photosynthesis or CO₂ fixation**, usually **one factor is the major cause** or is the one that **limits the rate**. Hence **at any point the rate will be determined by the factor available at sub-optimal levels**.
- **Blackman's (1905) Law of Limiting Factors** - comes into effect when several factors affect any [bio] chemical process. It states: **if a chemical process is affected by more than one factor, then its rate will be determined by the factor which is nearest to its minimal value; it is the factor which directly affects the process if its quantity is changed**.
- **For example**, despite the presence of a **green leaf and optimal light and CO₂ conditions**, the plant **may not photosynthesise** if the **temperature is very low**. This leaf, if given the **optimal temperature**, will start **photosynthesising**.

11.10.1

Light

We need to distinguish between **light quality, light intensity** and the **duration of exposure to light**, while discussing light as a factor that affects photosynthesis.

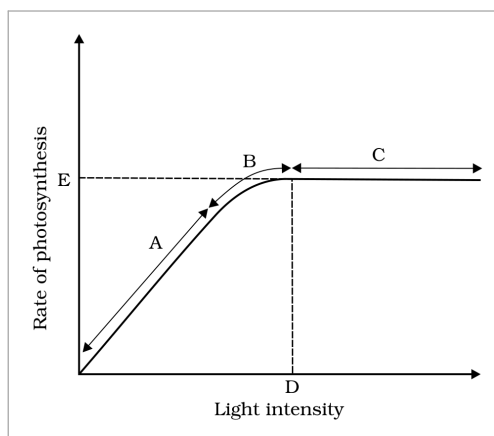


Fig. 11.10 - Graph of light intensity on the rate of photosynthesis.

The graph's vertical axis is the **Rate of photosynthesis** and its horizontal axis is **Light intensity**. The points **A, B, C, D** and **E** are marked along the curve.

- There is a **linear relationship** between incident light and CO_2 fixation rates at low light intensities.
- At **higher light intensities**, gradually the rate does not show further increase as other factors become limiting.
- **Light saturation** occurs at 10 per cent of the full sunlight. Hence, except for plants in shade or in dense forests, light is rarely a limiting factor in nature.
- Increase in incident light beyond a point causes the **breakdown of chlorophyll** and a decrease in photosynthesis.

11.10.2 Carbon dioxide Concentration

- Carbon dioxide is the major limiting factor for photosynthesis. The concentration of CO_2 is very low in the atmosphere (between 0.03 and 0.04 per cent).
- Increase in CO_2 concentration upto 0.05 per cent can cause an increase in CO_2 fixation rates; beyond this the levels can become damaging over longer periods.
- The C_3 and C_4 plants respond differently to CO_2 concentrations, and at low light conditions neither group responds to high CO_2 conditions. At high light intensities both C_3 and C_4 plants show increase in the rates of photosynthesis.

Plant group	Response to increased CO_2 concentration
C_4 plants	Show saturation at about 360 microlitre per litre
C_3 plants	Respond to increased CO_2 concentration; saturation is seen only beyond 450 microlitre per litre

- Thus, current availability of CO_2 levels is limiting to the C_3 plants.
- The fact that C_3 plants respond to higher CO_2 concentration by showing increased rates of photosynthesis leading to higher productivity has been used for some greenhouse crops such as tomatoes and bell pepper, which are allowed to grow in a carbon dioxide enriched atmosphere that leads to higher yields.

11.10.3 Temperature

- The dark reactions being enzymatic are temperature controlled. Though the light reactions are also temperature sensitive, they are affected to a much lesser extent.
- The C_4 plants respond to higher temperatures and show a higher rate of photosynthesis, while C_3 plants have a much lower temperature optimum.
- The temperature optimum for photosynthesis of different plants also depends on the habitat that they are adapted to: tropical plants have a higher temperature optimum than the plants adapted to temperate climates.

11.10.4 Water

- Even though water is one of the reactants in the light reaction, the effect of water as a factor is more through its effect on the plant, rather than directly on photosynthesis.
- Water stress causes the stomata to close, hence reducing the CO_2 availability.
- Besides, water stress also makes leaves wilt, thus reducing the surface area of the leaves and their metabolic activity as well.

Recap QUICK RECAP

- Green plants are autotrophs - they make their own food by photosynthesis, a physico-chemical process using light energy to synthesise organic compounds. CO_2 is taken in through stomata and used to make carbohydrates, principally glucose and starch.
- Photosynthesis occurs only in the green parts, mainly the leaves, whose mesophyll cells carry many chloroplasts. In the chloroplast, the membranes run the light reaction and the stroma runs the chemosynthetic (carbon-fixing) pathway - the two stages of photosynthesis.
- Four pigments separate on a chromatogram: chlorophyll a (the chief pigment), chlorophyll b, xanthophylls and carotenoids; the last three are accessory pigments, which widen the usable wavelength range and protect chlorophyll a from photo-oxidation. Most photosynthesis happens in the blue and red regions.

- Light energy absorbed by the **antenna** pigments is **funnelled to the reaction centre chlorophylls**. **PS I** has **P700**, **PS II** has **P680**. Electrons run **PS II to PS I to NADP⁺** along the **Z scheme**, giving **NADPH**; **splitting of water at PS II** supplies the replacement electrons and releases **O₂** and **protons**.
- **Non-cyclic photophosphorylation** (PS II then PS I) makes **ATP and NADPH**; **cyclic photophosphorylation** (PS I alone, in the **stroma lamellae**, which lack PS II and NADP reductase) makes **only ATP**. ATP synthesis is **chemiosmotic**: protons accumulate in the **thylakoid lumen**, and their return through **CF₀** drives a **conformational change in CF₁** of ATP synthase.
- In the **carbon fixation cycle**, **RuBisCO** adds **CO₂** to the 5-carbon **RuBP**, giving **2 molecules of 3-carbon PGA**. The **Calvin cycle - carboxylation, reduction, regeneration** - converts this to sugar and **regenerates RuBP**, at **3 ATP and 2 NADPH** per **CO₂**, so **6 CO₂, 18 ATP and 12 NADPH** yield **one glucose**.
- **RuBisCO** also catalyses a **wasteful oxygenation reaction in C₃ plants - photorespiration** - which makes **no sugar, no ATP and no NADPH**, and whose **biological function is not known yet**.
- **Some tropical plants** use the **C₄ pathway**. **PEPcase** fixes **CO₂** in the **mesophyll** to the **4-carbon OAA**; **C₄ acids** move to the **bundle sheath cells** ('Kranz' anatomy), where **decarboxylation** concentrates **CO₂** and the **Calvin pathway** makes the carbohydrate. This **abolishes photorespiration** and gives **higher productivity** and **tolerance of higher temperatures**.
- Rate-limiting factors follow **Blackman's Law of Limiting Factors**. **Light saturates at 10 per cent of full sunlight**; **CO₂ (0.03-0.04 per cent in air) is the major limiting factor**; **dark reactions are temperature controlled**; and **water acts indirectly, by closing stomata and wilting leaves**.

Appendix

TERMS USED IN THE EXERCISES

Four of the chapter's questions lean on something the chapter never states outright. Those four are reproduced here with their answers. The answers marked **[addition]** go beyond what this chapter states.

① **[NOTE]** C₃ and C₄ plants differ in leaf anatomy; C₄ leaves show Kranz anatomy, with bundle-sheath cells.

No. **[addition]** C₃ and C₄ plants **cannot be distinguished by external appearance**. Every criterion this chapter gives is **internal**: the '**Kranz**' anatomy - the **large bundle sheath cells** around the vascular bundles, with **many chloroplasts, thick walls impervious to gaseous exchange** and **no intercellular spaces** - which is visible **only in a section of the leaf**, not from the outside.

① **[NOTE]** Chlorophyll a is the chief pigment of photosynthesis. Accessory pigments (chlorophyll b, xanthophylls and carotenoids) absorb a wider range of wavelengths and protect chlorophyll a from photo-oxidation.

No, it would not. **[addition]** **Chlorophyll a is indispensable**: this chapter states that chlorophyll a is the **chief pigment associated with photosynthesis**, and that the **single chlorophyll a molecule forms the reaction centre** of each photosystem. A plant with **only chlorophyll b** would have **no reaction centre**, so the **light energy it absorbed could not be converted**. Chlorophyll b, xanthophylls and carotenoids are **accessory pigments**: they **absorb light and transfer the energy to chlorophyll a**, which both **widens the range of wavelengths** usable for photosynthesis and **protects chlorophyll a from photo-oxidation**.

☆ **[MEMORY AID - not in NCERT]** A leaf kept in the dark turns yellow or pale green because chlorophyll breaks down while the more stable carotenoids and xanthophylls remain.

[addition] A leaf kept in the dark turns **yellow or pale green** because **chlorophyll breaks down faster in the dark than the carotenoids do**. As the green chlorophyll is lost, the **yellow accessory pigments - the xanthophylls and carotenoids - are left behind** and their colour shows through. The **more stable pigments are therefore the carotenoids** (and xanthophylls); **chlorophyll is the less stable**. This chapter states that the accessory pigments **protect chlorophyll a from photo-oxidation**, but it does not give this dark-degradation ordering.

The **leaves on the shady side** - and the **potted plants kept in the shade** - have the **darker green** leaves. **[addition]** A **shade leaf builds more chlorophyll per unit leaf area**, so that it can **trap as much as possible of the little light it receives**; the extra chlorophyll is what makes it look darker green. This chapter states only that **light is rarely a limiting factor in nature except for plants in shade or in dense forests**, and that **increase in incident light beyond a point causes the breakdown of chlorophyll** - so the sunny-side leaf keeps less chlorophyll.

Note

A Note on the PS II / PS I Electron Replacement Wording

① **[NOTE]** **True fact**: the electron flow through the light reaction is one continuous, non-cyclic chain - **water -> PS II -> PS I -> NADP⁺**. Water splitting at PS II is the ultimate source of every electron in the chain, so it is water's electrons that end up replacing what PS I loses, using PS II purely as the relay in between.

① **[NOTE]** As **NCERT itself states it** (Ch.11, SS11.6.1, verbatim): "The electrons that were moved from photosystem II must be replaced. This is achieved by electrons available due to splitting of water... The electrons needed to replace those removed from photosystem I are provided by photosystem II." NCERT describes this as two separate replacement steps - water replaces PS II's electrons, then PS II replaces PS I's electrons - rather than as one continuous chain, which is why a first read can suggest PS II's electrons are being replaced twice. This is a known quirk of NCERT's own phrasing, carried here exactly as printed, not an error introduced in these notes.